Quiz: Dynamo

Test your knowledge of Dynamo by completing this quiz.

Quiz ?

Statement

In CAP theorem terms, Dynamo is ___ system?

Select Multiple Options

☐ AP (Available and Partition Tolerant)

☐ CP (Consistent and Partition Tolerant)

☐ CA (Consistent and Available)

Check Answer

Quiz ?

Statement

Amazon Dynamo is called a zero-hop DHT because:

Select Multiple Options

☐ A collision-free hash function is used.

☐ Consistent hashing allows the system to grow.

☐ Clients know the complete set of servers and can directly contact the server that holds the data they need.

☐ Each server node maintains a finger table that identifies other nodes.

Check Answer

Quiz ?

Statement

Does Dynamo always write data to the first 'N' nodes in the preference list?

Select Option

○ True

○ False

Check Answer

Quiz ?

Statement

How does Dynamo resolve conflicts between different versions of data?

Select Multiple Options

☐ Merkle tree

☐ Hinted handoff

☐ Vector clock

☐ Gossip protocol

Check Answer

Quiz ?

Statement

What strategies can be used to resolve data conflicts? Select all that apply.

Select Multiple Options

☐ Last write wins

☐ Conflict-free replicated data types (CRDTs).

☐ Sloppy quorum

☐ Vector clocks

Check Answer

Quiz ?

Statement

How does Dynamo handle temporary node failures? Select all that apply.

Select Multiple Options

☐ Merkle tree

☐ Hinted handoff

☐ Sloppy quorum

☐ Gossip protocol

Check Answer

Quiz ?

Statement

What are the benefits of Vnodes in Dynamo? Select all that apply.

Select Multiple Options

☐ Vnodes help spread the load more evenly across the physical nodes on the cluster. This speeds up the rebalancing process after adding or removing nodes.

☐ Vnodes enable the use of virtual machines to help Dynamo scale better.

☐ Vnodes make it easier to maintain a cluster containing heterogeneous machines.

☐ Use of Vnodes decreases the probability of hotspots, as Vnodes help assign smaller ranges to each physical node.

Check Answer

Quiz ?

Statement

What is optimistic replication in Dynamo?

Select Multiple Options

☐ When replicating, Dynamo chooses the closest nodes to the coordinator node to store the replicas.

☐ Dynamo tries to distribute replicas on different racks.

☐ Replicas may diverge and are not guaranteed to be identical at all times.

☐ All replicas are updated atomically, ensuring ACID semantics.

Check Answer

Quiz ?

Statement

Dynamo uses Vector clocks to:

Select Multiple Options

☐ Impose a total ordering of all get and put operations in the system.

☐ Disallow concurrent updates of the same object by choosing the earliest writer.

☐ Determine the sequence of write operations across multiple objects.

☐ Identify conflicting versions of the same object stored on different servers.

Check Answer

Quiz ?

Statement

Dynamo optimizes key lookups via:

Select Multiple Options

☐ Consistent hashing

☐ Finger tables

☐ Storing the table of all nodes in the system at each node.

☐ Storing all key-value sets in a hash table at each node.

Check Answer

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☐ Mark as Completed